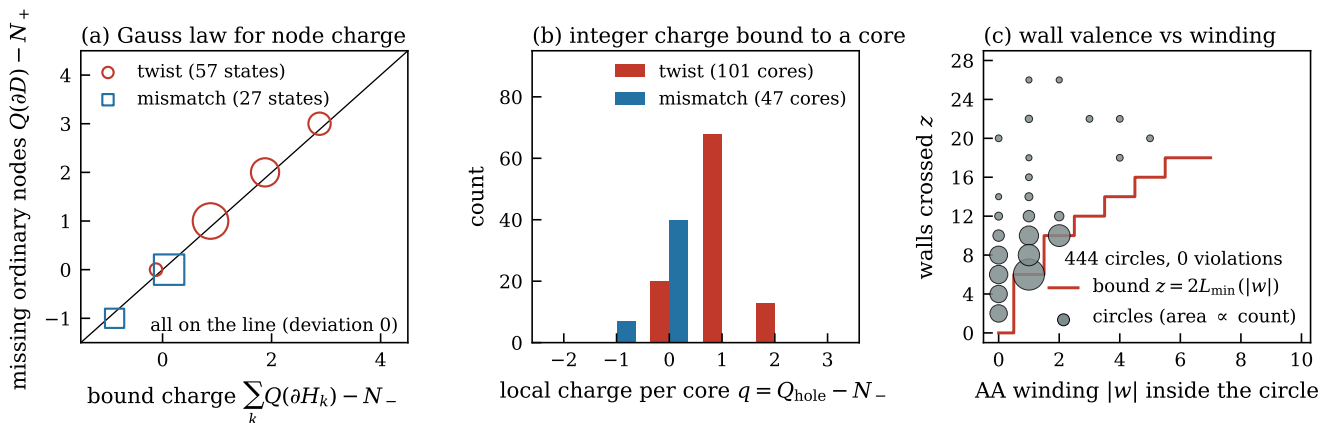
(a) Gauss law for node charge
twist (57 states)
mismatch (27 states)
all on the line (deviation 0)
missing ordinary nodes $Q(\partial D) - N_+$
bound charge $\sum_k Q(\partial H_k) - N_-$
(b) integer charge bound to a core
twist (101 cores)
mismatch (47 cores)
count
local charge per core $q = Q_{\text{hole}} - N_-$
(c) wall valence vs winding
444 circles, 0 violations
bound $z = 2L_{\min}(|w|)$
circles (area $\propto$ count)
walls crossed z
AA winding $|w|$ inside the circle